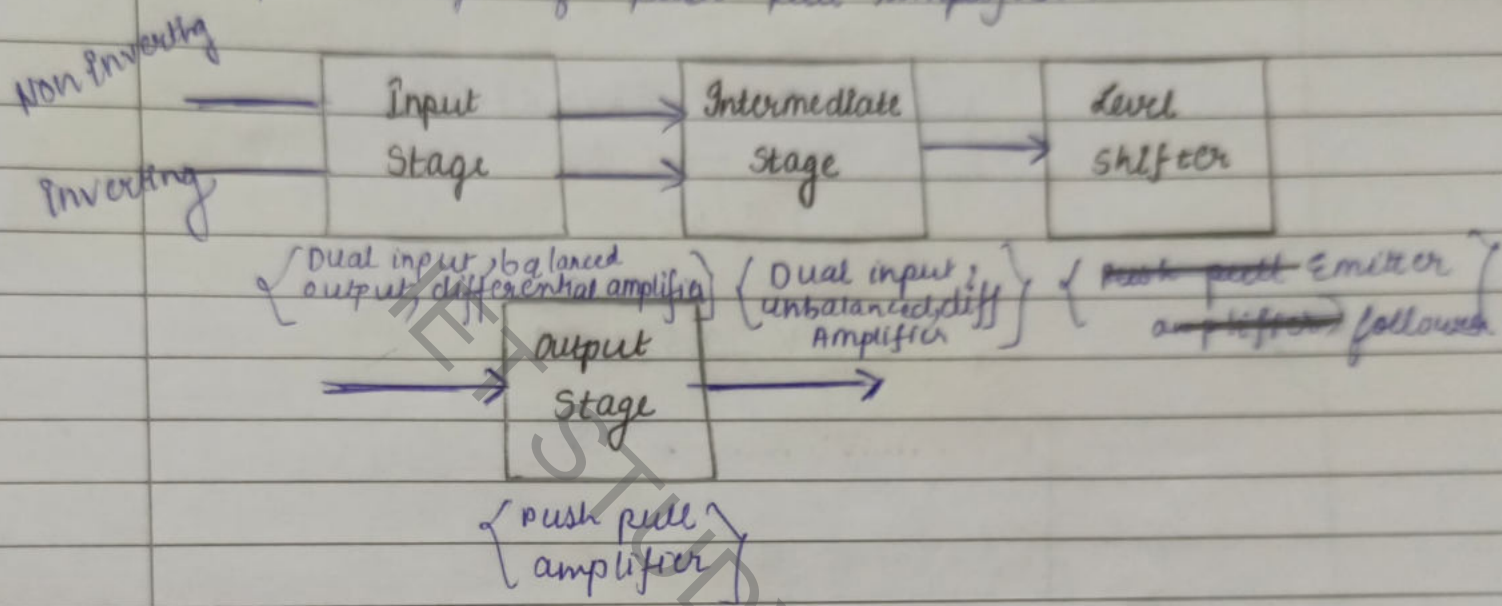
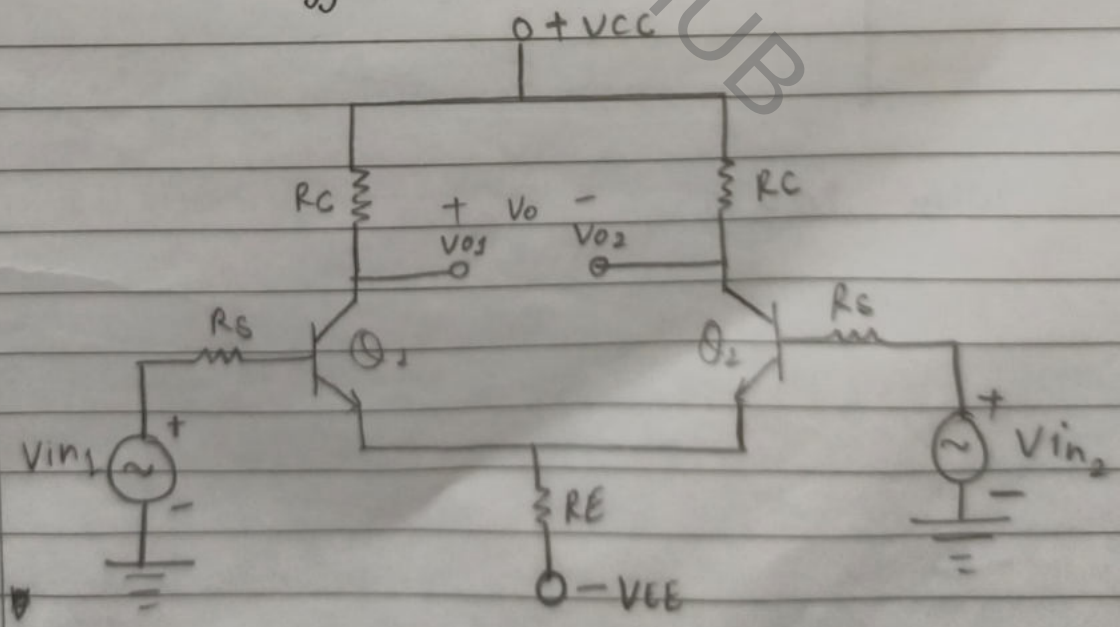


Operational Amplifier (OP-AMP)

An operational amplifier is a direct coupled high gain amplifier usually consisting of one or more differential amplifier followed by level shifter and output stage of push pull amplifier.



Differential Amplifier



$$V_O = V_{O1} - V_{O2}$$

Mode of operation

* Differential mode

$$V_d = V_{in1} - V_{in2}$$

$$A_d = \frac{V_o}{V_d}$$

* Common mode

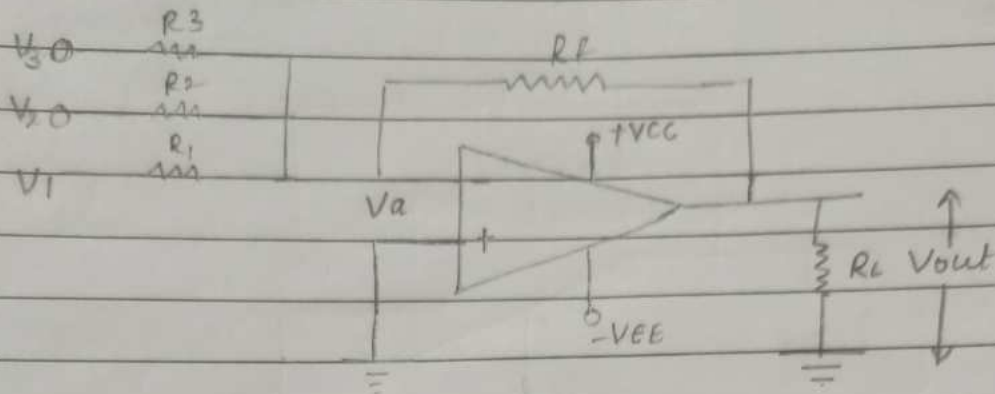
$$V_{cm} = \frac{V_{in1} + V_{in2}}{2}$$

$$A_{cm} = \frac{V_o}{V_{cm}}$$

$$V_o = V_{o1} - V_{o2}$$

$$V_o = A_d V_d + A_{cm} V_{cm}$$

OP - AMP as summing amplifier



$$\frac{V_a - V_1}{R_1} + \frac{V_a - V_2}{R_2} + \frac{V_a - V_3}{R_3} + \frac{V_a - V_{out}}{R_f} = 0$$

$$V_a = 0$$

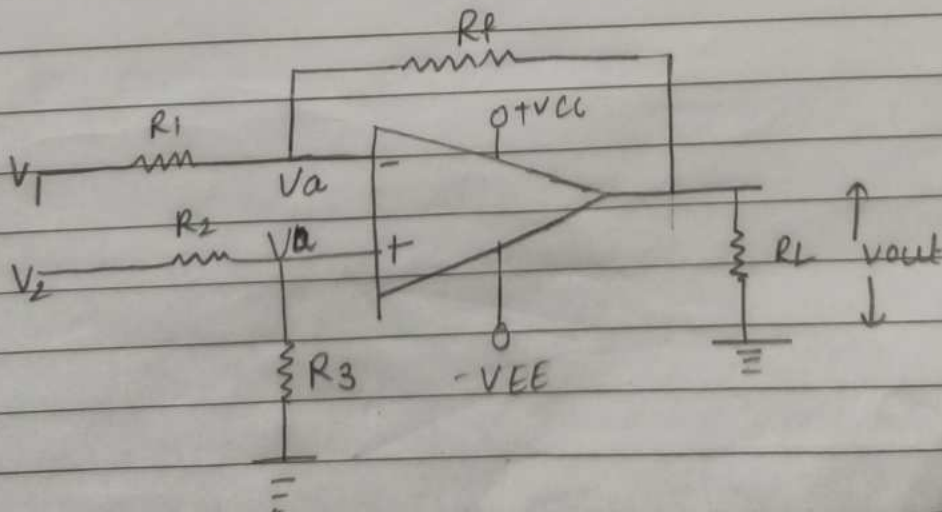
$$-\frac{V_1}{R_1} - \frac{V_2}{R_2} - \frac{V_3}{R_3} = \frac{V_{out}}{R_f}$$

$$V_{out} = -R_f \left(\frac{V_1}{R_1} + \frac{V_2}{R_2} + \frac{V_3}{R_3} \right)$$

$$R_1 = R_2 = R_3 = R$$

$$V_{out} = -\frac{R_f}{R} (V_1 + V_2 + V_3)$$

OP - AMP as subtractor (Difference Amplifier)



$$\frac{V_a - V_1}{R_1} + \frac{V_a - V_{out}}{R_f} = 0$$

$$\frac{V_a - V_2}{R_2} + \frac{V_a - 0}{R_3} = 0$$

$$\frac{V_a}{R_2} + \frac{V_a}{R_3} = \frac{V_2}{R_2}$$

$$V_a \left(\frac{R_2 + R_3}{R_2 R_3} \right) = \frac{V_2}{R_2}$$

$$V_a = \frac{V_2 R_3}{R_2 + R_3}$$

$$\frac{\left(\frac{V_2 R_3}{R_2 + R_3} \right) - V_1}{R_1} + \frac{\left(\frac{V_2 R_3}{R_2 + R_3} \right) - V_{out}}{R_f} = 0$$

$$\frac{V_2 R_3}{R_1(R_2 + R_3)} - \frac{V_1}{R_1} + \frac{V_2 R_3}{R_f(R_2 + R_3)} - \frac{V_{out}}{R_f} = 0$$

$$R_1 = R_2 = R_3 = R$$

$$\frac{\frac{V_2 R}{2R} - V_1}{R} + \frac{\frac{V_2 R}{2R} - V_{out}}{R_f} = 0$$

$$\frac{\frac{V_2}{2} - \frac{V_1}{R}}{R} + \frac{\frac{V_2}{2} - \frac{V_{out}}{R_f}}{R_f} = 0$$

$$\frac{V_2 - 2V_1}{2R} + \frac{V_2 - 2V_{out}}{2R_f} = 0$$

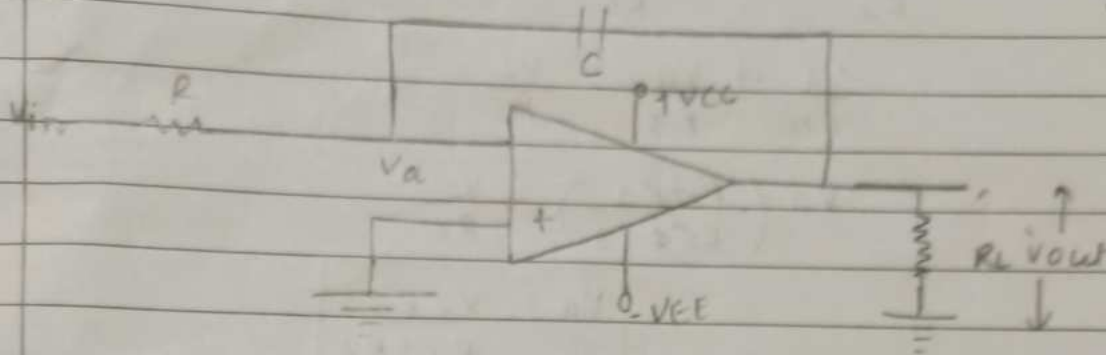
$$R_f = R$$

$$2V_2 - 2V_1 - 2V_{out} = 0$$

$$V_{out} = V_1 + V_2$$

OP AMP as an integrator

$$I_c = C \frac{dV_c}{dt}$$



$$\frac{V_a - V_{in}}{R} + C \frac{dV_a - V_{out}}{dt} = 0$$

$$V_a = 0 \text{ (Virtual ground concept)}$$

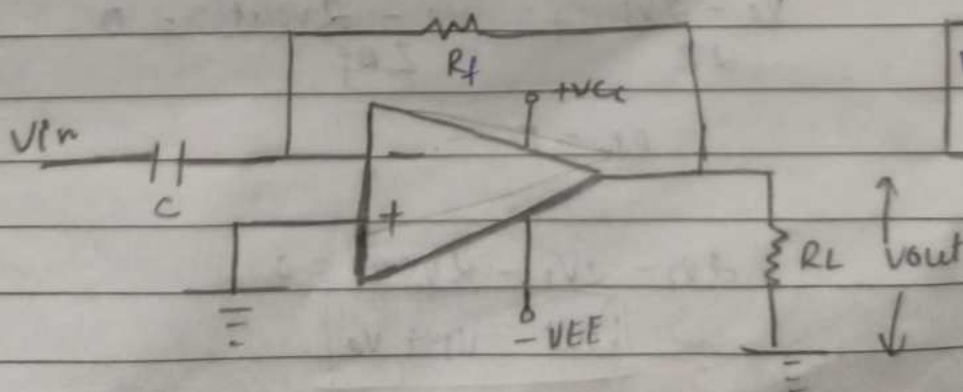
$$-\frac{V_{in}}{R} + C \frac{d(-V_{out})}{dt} = 0$$

$$-C \frac{dV_{out}}{dt} = \frac{V_{in}}{R}$$

$$\frac{dV_{out}}{dt} = -\frac{1}{RC} V_{in}$$

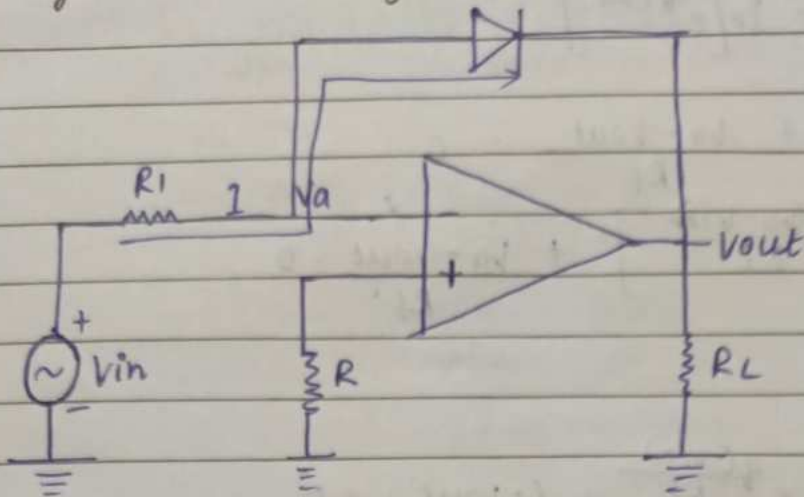
$$V_{out} = -\frac{1}{RC} \int V_{in} dt$$

OP AMP as a differentiator



$$V_{out} = -R_f C \frac{dV_{in}}{dt}$$

Logarithmic amplifier



$$I = I_0 \left[e^{qV/KT} - 1 \right] \quad e^{qV/KT} \gg 1$$

$$I = I_0 \left[e^{qV/KT} \right]$$

$$\frac{V_a - V_{in}}{R_i} + I_0 \left[e^{q \frac{(V_a - V_{out})}{KT}} \right] = 0$$

$$V_a = 0 \text{ (virtual ground)}$$

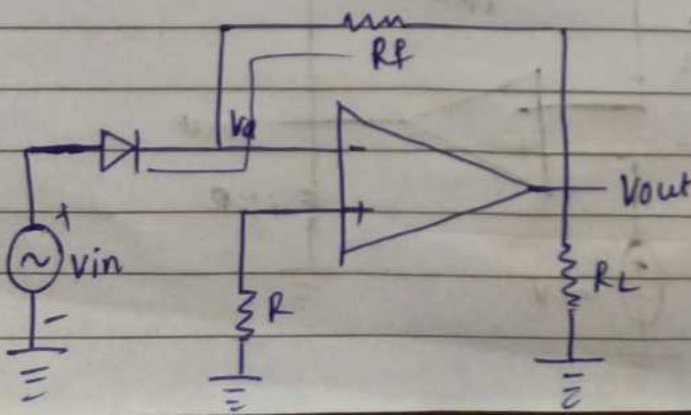
$$\frac{-V_{in}}{R_i} = -I_0 \left[e^{q \frac{(-V_{out})}{KT}} \right]$$

$$e^{-\frac{qV_{out}}{KT}} = \frac{V_{in}}{R_i I_0}$$

$$-\frac{qV_{out}}{KT} = \ln \left(\frac{V_{in}}{R_i I_0} \right)$$

$$V_{out} = -\frac{KT}{q} \ln \left(\frac{V_{in}}{R_i I_0} \right)$$

Antilogarithmic amplifier



$$I = I_0 \left[e^{\frac{qV}{kT}} \right]$$

$$I + \frac{V_a - V_{out}}{R_f} = 0$$

$$I_0 \left[e^{\frac{q(V_a - V_{in})}{kT}} \right] + \frac{V_a - V_{out}}{R_f} = 0$$

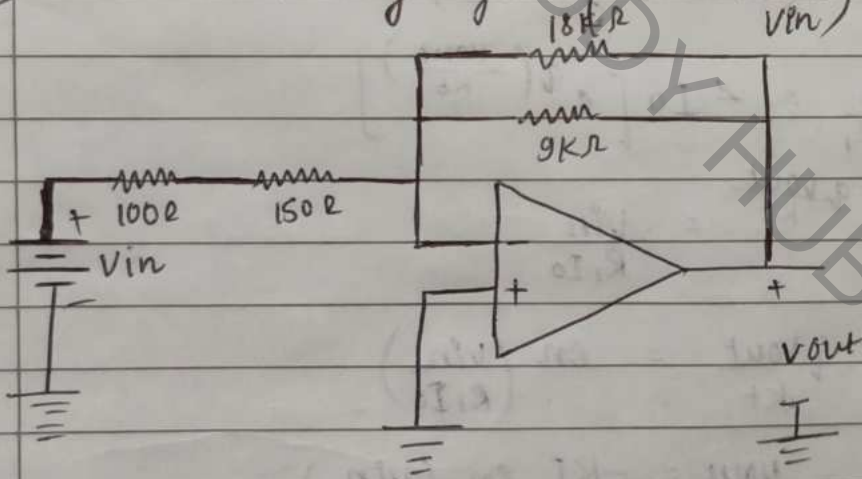
$$V_a = 0$$

$$I_0 e^{\frac{q(-V_{in})}{kT}} + \left(\frac{-V_{out}}{R_f} \right) = 0$$

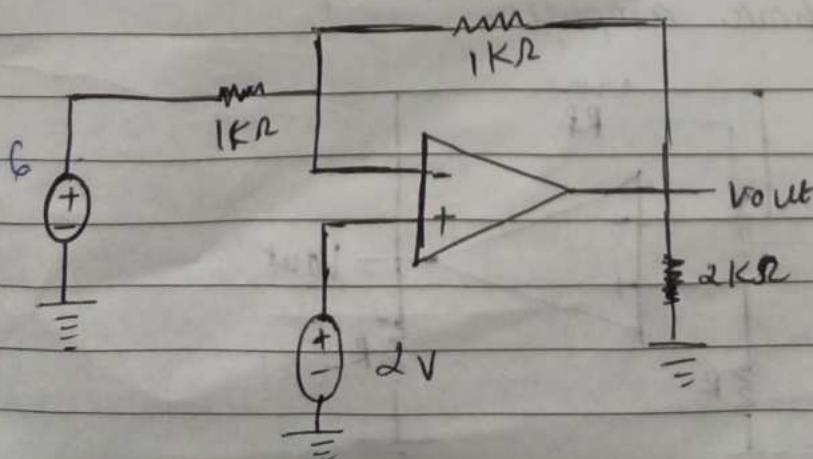
$$I_0 e^{-\frac{qV_{in}}{kT}} = \frac{V_{out}}{R_f}$$

$$V_{out} = R_f I_0 e^{-\frac{qV_{in}}{kT}}$$

Ques 1 Find out voltage gain ($A_v = \frac{V_{out}}{V_{in}}$)



Ques 2 Find out output voltage (V_{out})



$$\frac{V_a - V_{in}}{250} + \frac{V_a - V_{out}}{6000} = 0$$

$$V_a = 0$$

$$\frac{-V_{in}}{250} - \frac{V_{out}}{6000} = 0$$

$$\frac{-V_{in}}{250} = \frac{V_{out}}{6000}$$

$$\frac{-6000}{250} = \frac{V_{out}}{V_{in}}$$

$$\frac{V_{out}}{V_{in}} = -24$$

$$\frac{V_a - 6}{1000} + \frac{V_a - V_{out}}{1000}$$